

Project Progress

Nowcasting Inflation via Daily Web Scraping Price

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1. Objective

- Build an Automated Data Infrastructure and Monitoring System:** Set up an automated data pipeline and database to collect, store, and continuously monitor prices.
- Calculate CPI for Inflation Nowcasting Using Machine Learning Model:** Apply an economic formula to compute daily price relatives, and use regularized Ridge regression modeling to nowcast inflation ahead of the official monthly release.
- Build a Decision-Support Dashboard:** Develop an interactive dashboard that presents daily CPI results and inflation for easy monitoring.

2. Architecture

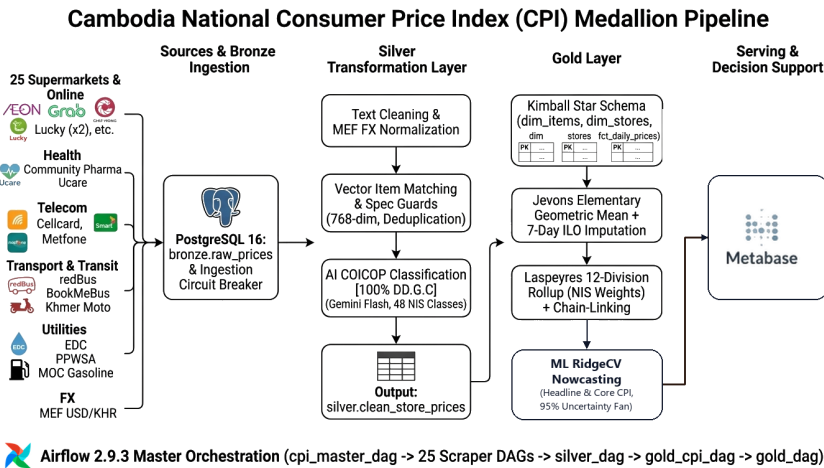


Figure 1: End-to-End System Architecture: Web Scraping, Medallion Pipeline & Dashboard.

Technology Stack

Component / Pipeline Layer	Technology Stack
Data Ingestion & Scraping	Python
Database and Storage	PostgreSQL 16
Orchestration	Apache Airflow
Data Transformation	dbt Core
AI Product Classifications	Gemini AI
Dashboard and Monitoring	Metabase
System Container	Docker

3. Data Collection

Data Source	Description
AEON Online Cambodia	Supermarket
DeliShop Asia Phnom Penh	Supermarket
GrabMart (Lucky Supermarket, Chip Mong)	Supermarket
L192 Online Marketplace	Footwear & Homeware
Community Pharmacy & Ucare Pharmacy	Healthcare & Pharmaceutical Goods
Khmer24 & Realestate.com.kh	Urban Residential Apartment, House Rentals
redBus & BookMeBus Transit	Bus & Travel Transport
Cellcard, Smart Axiata, Metfone	Mobile Plan and Wi-Fi
Tela Khmer	Gasoline
dataEF API	Exchange Rate
Sokha, Hyatt, Bayon BKK	Restaurant and Hotel

4. Methodology

Stage 1: Web Scraping Layer

Automated Daily Ingestion: Every morning at 08:00 AM, Apache Airflow orchestrates daily automated data extraction across 11 key retail and service sectors in Cambodia.

Protocols & Methods:

- **Direct Store APIs (REST & GraphQL):** Directly targets internal reverse-proxy endpoints (AEON 1, AEON 3, DeliShop Asia, BookMeBus) and GraphQL queries (L192) to download structured JSON product catalogs, bypassing client-side web rendering.
- **TLS Fingerprinting (curl_cffi):** Executes HTTP requests with Chrome TLS/JA3 fingerprint impersonation to bypass Cloudflare anti-bot defenses and Web Application Firewalls (WAF) with polite throttling (0.5s–1.0s delay).
- **Dynamic Browser Escalation (Headless Playwright):** Automatically launches an asynchronous headless Chromium browser context when encountering Cloudflare challenges or HTTP 403 Forbidden barriers (e.g., AEON) to render dynamic JavaScript and extract live prices.
- **Public Web Feeds & HTML Parsing:** Uses BeautifulSoup to extract housing rentals (Khmer24, Realestate.com.kh), scrapes official fuel price caps via Tela Khmer Telegram web feeds with Khmer numeral translation (Khmer digits → 0–9), and fetches daily official MEF USD/KHR exchange rates.
- **Defensive Quality Controls:** Enforces a zero-record circuit breaker (raises an error on empty scrapes to prevent data wipeout) and maintains a static fallback matrix if third-party websites undergo outages.

Stage 2: Bronze Layer (Raw Lake Ingestion)

Database Table: bronze.raw_prices

- **What it does:** Acts as an immutable, append-only raw data repository preserving the exact raw data as fetched from the web.
- **Stored Fields:** store_id, source_name, item_description_raw, price, currency, source_url, scraped_at, batch_id.

Stage 3: Silver Layer (Standardization, AI Classification & Quality Control)

Database Table: silver.standardized_prices

The Silver Layer has 3 main jobs:

[Raw Scrape] → 1. Clean Data → 2. Match Product → 3. Classify Basket → [Clean Silver Item]

1. Clean the Data (Standardize)

- **Removes noise:** Deletes marketing words (SALE, HOT DEAL, 50% OFF) and old price tags.
- **Normalizes units:** Converts volumes and weights into standard sizes (500ml, 1kg) to calculate price per liter/kg.
- **Handles dual currency:** Converts USD and Khmer Riel using the daily MEF exchange rate.
- **Fixes Khmer text:** Segments unspaced Khmer words and translates digits (Khmer digits → 0–9).

2. Match the Product (Find the Same Item)

Connects identical products across different stores (e.g., AEON vs. DeliShop) using a 5-step waterfall:

- **Barcode (GTIN):** Exact match.
- **Store SKU:** Match with past days’ catalog.
- **Clean Name & Size:** Exact text match with size protection (prevents 330ml matching 1.5L).
- **AI Vector (pgvector):** Matches English and Khmer equivalents (≥ 95% similarity).
- **Fuzzy Spelling:** Catches small typos.

3. Classify into Official Baskets (UN COICOP)

- Uses Google Gemini AI in batches to assign the official 4-digit government code (e.g., 01.1.1.1 for Rice/Bread).
- **Strict Guardrails prevent AI mistakes:**
 - Pet Food → Goes to Pets (09.3.1), never human food.
 - Shampoo / Soap → Goes to Personal Care (12.1.3), never food.
 - Beer / Wine → Goes to Alcohol (02.1), isolated from soft drinks.

The Gold Layer in Brief: From Clean Products to Official CPI

The Gold Layer applies official international standards (ILO / UN guidelines) to turn thousands of clean daily prices into national economic indicators in 3 main steps:

[Clean Silver Items] → 1. Jevons Index → 2. Laspeyres Aggregation → 3. Core CPI & Marts

1. Elementary Index Calculation (Jevons Formula)

- **Jevons Geometric Mean:** Instead of a simple average (which gets distorted by expensive luxury items), the system uses the geometric mean:

$$I_{\text{rice},t} = \left(\frac{P_{1,t}}{P_{1,0}} \times \frac{P_{2,t}}{P_{2,0}} \times \dots \times \frac{P_{n,t}}{P_{n,0}} \right)^{\frac{1}{n}}$$

- **Missing Price Imputation:** If an item is temporarily out-of-stock, a 7-day carry-forward rule fills in the price so the index doesn’t jump wildly.

2. Official National Aggregation (Laspeyres Formula)

Cambodians spend more money on food than on furniture. To reflect real life, the 12 COICOP divisions are combined using official government survey expenditure weights (CSES 2020):

$$\text{Headline CPI} = \sum \left(\text{Weight} \times \text{Division Index} \right)$$

- **Food & Drinks (01):** 44.775% (Highest weight in Cambodia)
- **Housing & Electricity (04):** 17.062%
- **Transport & Fuel (07):** 12.203%
- **All other 9 Divisions:** Remaining 25.960% (Clothing, Health, Telecom, etc.)

3. Central Bank Metric: Core CPI

For central bankers at the National Bank of Cambodia (NBC), temporary global oil spikes or seasonal vegetable floods distort the real picture.

- **Refined Core CPI:** The system calculates a second official index that removes Food (01) and Energy (04):

$$\text{Core CPI} = \frac{\sum_{\text{ex-Food, ex-Energy}} w_j \cdot I_j}{\sum w_j}$$

This shows the underlying, long-term price stability of the economy.

Stage 4: Nowcasting

- **Methodology (Two-Tier Hybrid Approach):** Rather than using a black-box AI model, we use a hybrid economic approach:
 - *Tier 1 (Official Aggregation):* We use fixed CSES expenditure weights across all 12 COICOP divisions following official NIS/ILO Laspeyres standards. We calculate both Headline CPI and Core CPI (excluding volatile food and energy).
 - *Tier 2 (Forward Econometric Projection):* For past observed days, we compute the realized daily average from scraped prices. For the unobserved remaining days of the month, we nowcast daily drift rates using regularized Ridge Regression (RidgeCV) with Bayesian prior shrinkage.
 - *Blending:* We combine realized days and projected days using a time-weighted formula that smoothly converges onto actual data as the month ends.
- **Cambodia-Specific Market Features:**
 - *5 Core Baskets:* We model the 5 high-velocity categories (~81.6% of Cambodia's basket): Food (44.8%), Housing/Utilities (17.1%), Transport (12.2%), Restaurants (3.1%), and Alcohol (1.6%).
 - *Freight Spillover:* Fuel price momentum (Transport) is selectively fed into Food and Restaurant catering to model logistics cost pass-through.
 - *Holiday Handling:* We built an exponential decay kernel for Khmer holidays (Khmer New Year, Pchum Ben, Water Festival) so temporary ceremonial spikes are not misread as permanent inflation.
 - *FX Pass-Through:* Ingests daily USD/KHR commercial exchange rates.

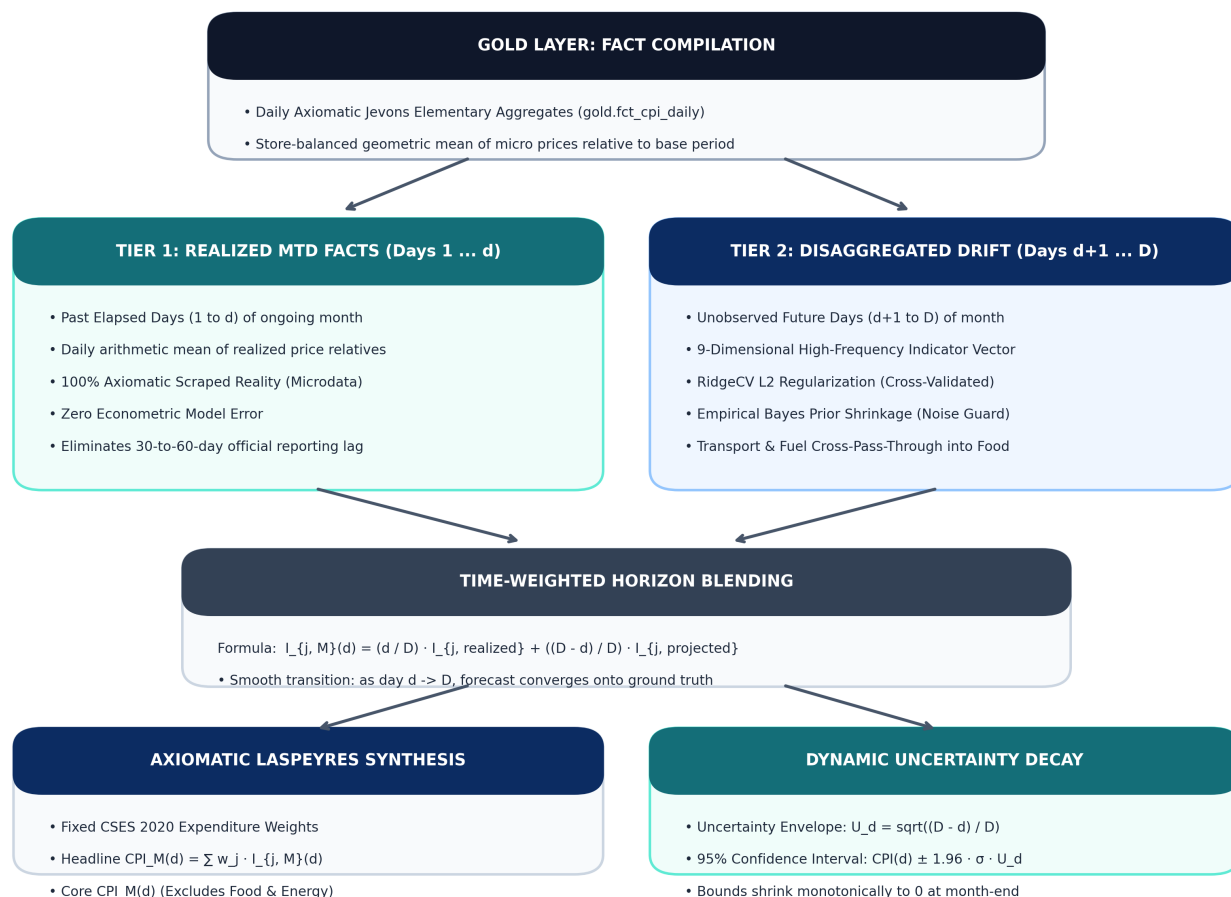


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Figure 2: Architecture Flowchart of the Two-Tier Inflation Nowcasting Engine

Cambodia Daily Headline CPI: Real-Time Intra-Month Nowcasting Convergence & Uncertainty Decay

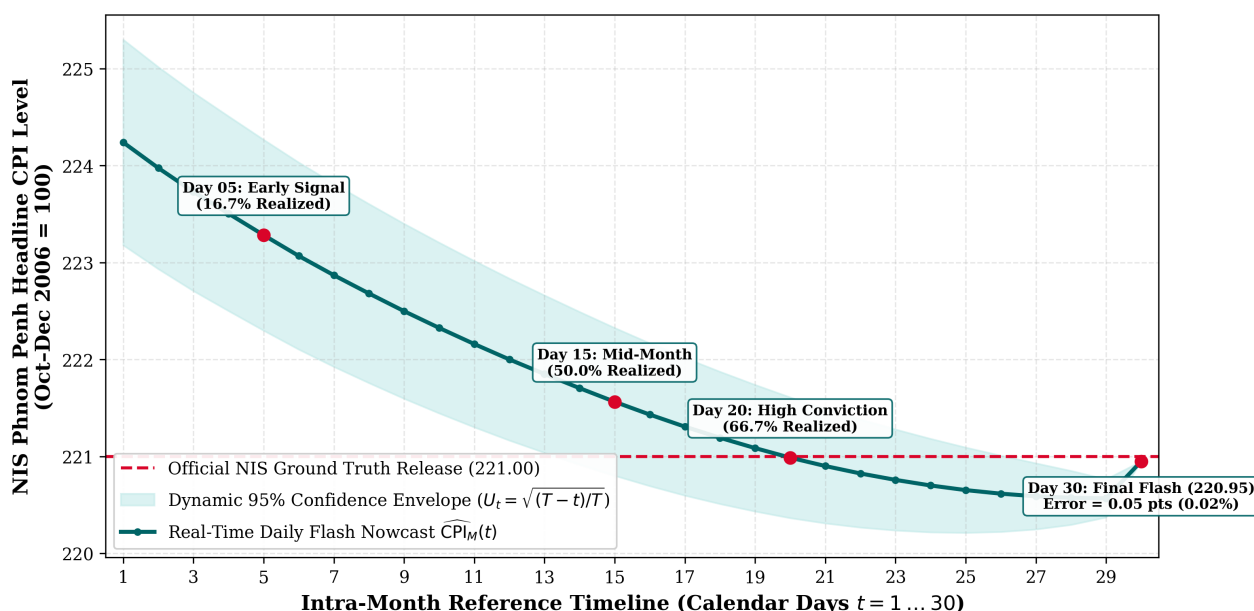


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Figure 3: Intra-Month Nowcasting Convergence with Monotonically Decaying Error Bounds

5. Decision-Support Metabase Dashboards & Monitoring

To provide actionable macroeconomic intelligence and continuous pipeline observability, the system implements three dedicated production dashboards in Metabase (deployed via Docker containers):

Dashboard 1: Macro CPI & Inflation Analytics

Presents real-time daily Headline CPI and Core CPI (excluding volatile Food & Energy), month-over-month (MoM) and year-over-year (YoY) inflation rates, 12 COICOP division breakdowns, and out-of-sample tracking error against official National Institute of Statistics (NIS) monthly reports.

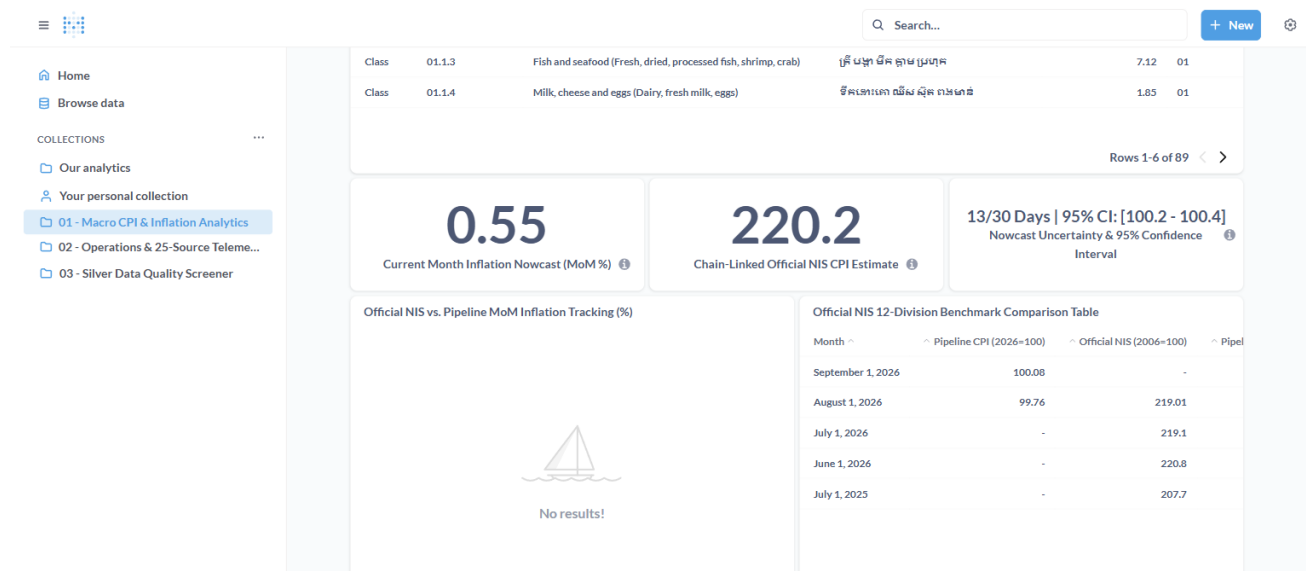


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Figure 4: Metabase Dashboard 1 — Real-Time Macro CPI & Inflation Analytics

Dashboard 2: Operations & Data Source Telemetry

Monitors daily ingestion health across all 11 retail and service data sources. Displays daily scraped observation counts, scraper execution runtimes, HTTP status codes, browser automation latency, and active catalog availability.

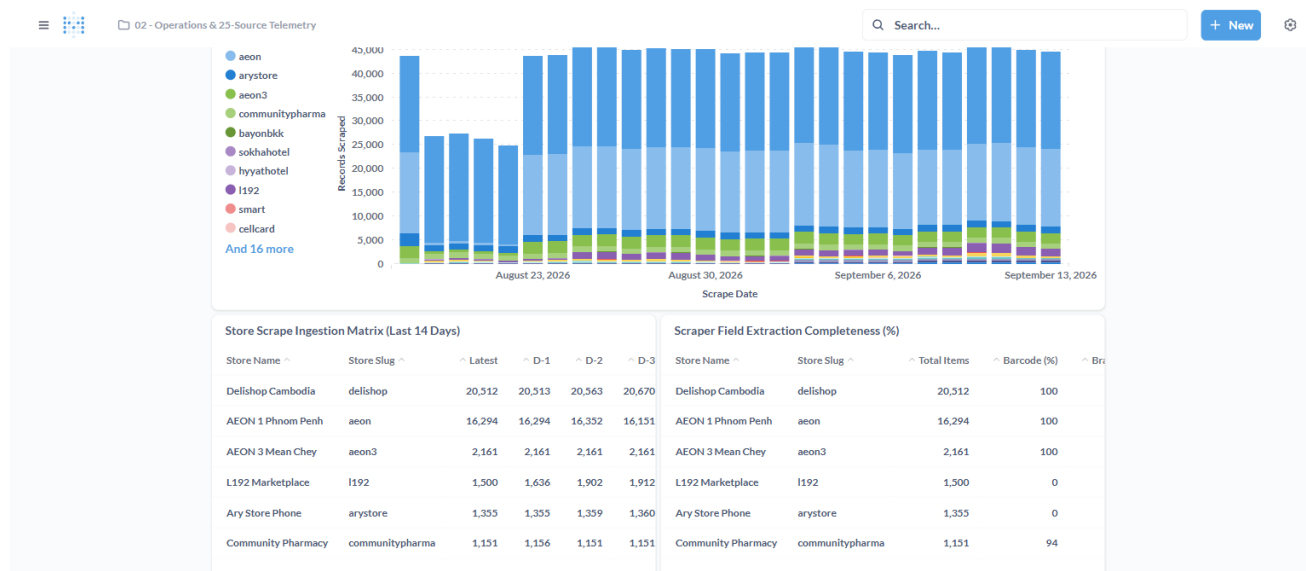


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Figure 5: Metabase Dashboard 2 — Data Lake Operations & Scraping Telemetry

Dashboard 3: Silver Layer Data Quality Screener

Performs live quality screening across raw-to-silver transformations. Tracks unit normalization rates (conversions to 1kg/1L), dual currency conversion (USD/KHR via daily MEF rates), 5-step entity matching waterfall hit rates (Barcode, SKU, text+size, vector pgvector, fuzzy), and Gemini AI classification guardrails (preventing pet food/cleaning chemical classification errors).

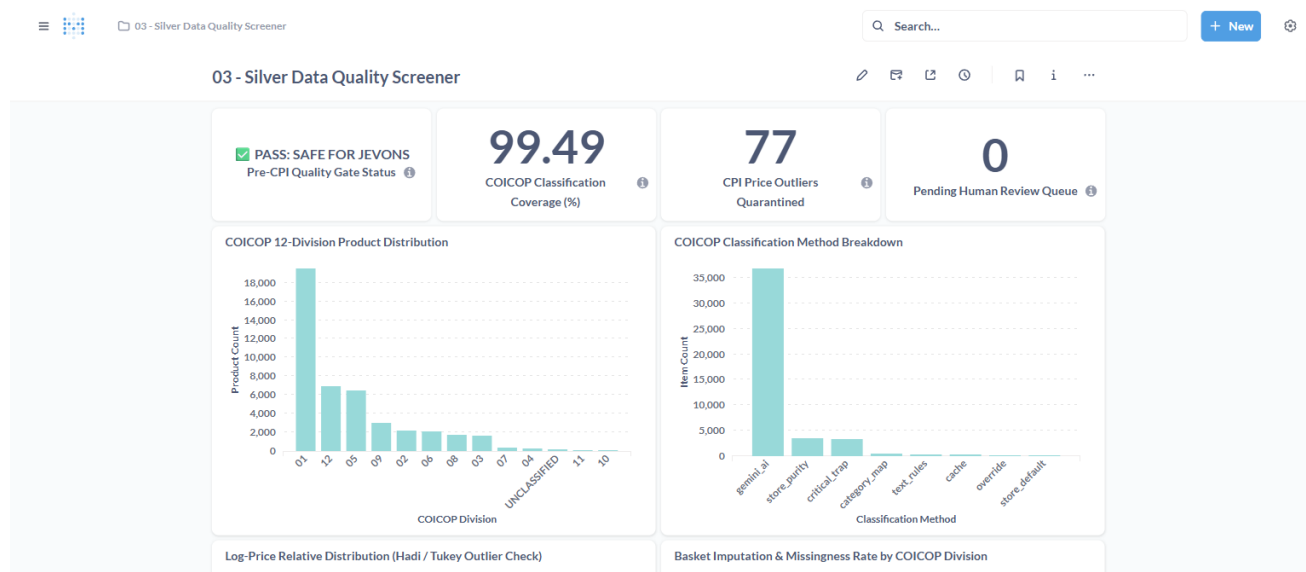


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Figure 6: Metabase Dashboard 3 — Silver Layer Data Quality & Classification Screener